Question 1: Multi-Stage Cepstral Feature Extraction & Phoneme Boundary Detection

Audio: heisenberg.mp3 | Sample Rate: 16 kHz

1. Manual MFCC / Cepstrum Engine

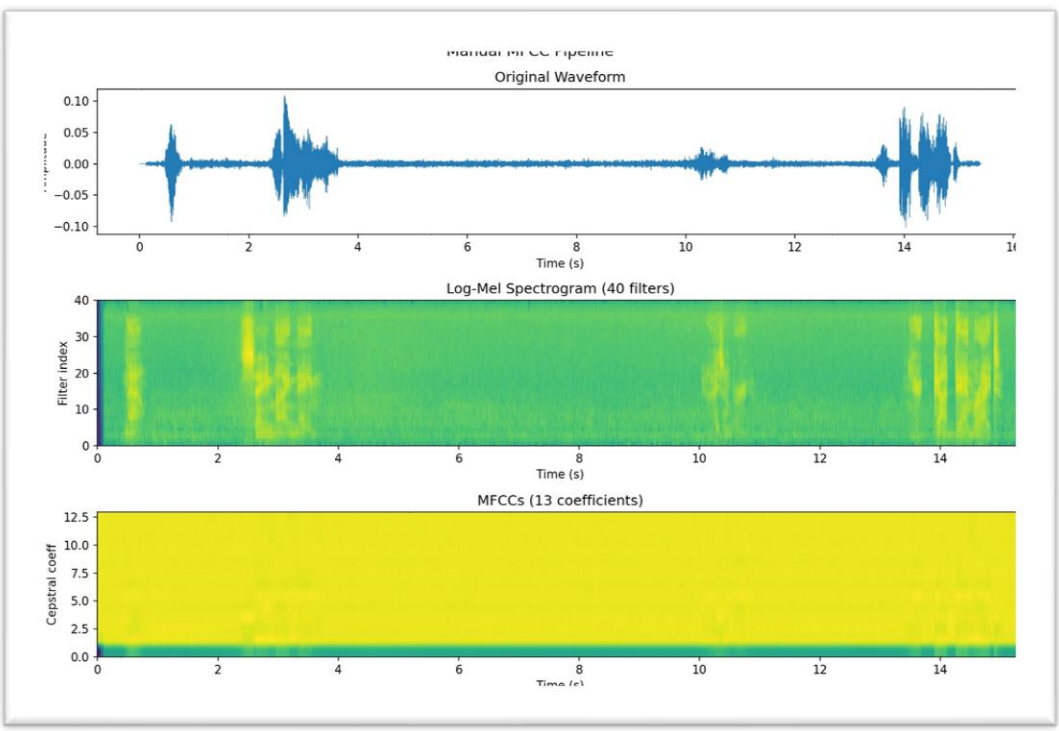
In this experiment, the MFCC pipeline was implemented from scratch instead of using built-in libraries to understand each step.

1.1 Pipeline Steps & Key Hyperparameters

- Pre-emphasis: y[n] = x[n] - 0.97x[n-1]
- Framing & Windowing: 25 ms frames with 10 ms overlap using Hamming window
- FFT & Mel Filterbank: 512 FFT and 40 Mel filters
- Log + DCT: First 13 coefficients taken as MFCC
- The configuration uses a pre-emphasis α of 0.97, a frame length of 25 ms with a 10 ms frame hop, a Hamming window, an NFFT of 512, 40 Mel filters, and a final MFCC count of 13.

1.2 Output

An MFCC matrix of shape (1502, 13).



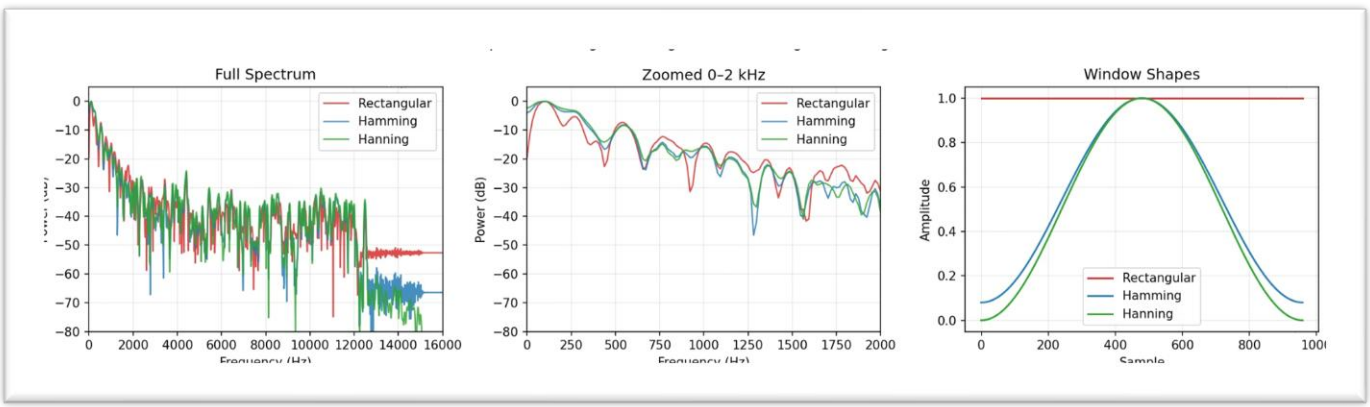
2. Spectral Leakage & SNR Analysis

A 30 ms frame was extracted starting at t = 100 ms and multiplied by three window functions. For each window, the SNR, spectral leakage and main-lobe −3 dB bandwidth were computed.

2.1 Results Table

Window	Leakage Ratio	SNR (dB)	Main Lobe Width (bins)
Rectangular	0.9980	−3.30	~2 (narrow but high sidelobes)
Hamming	0.9975	−1.70	~4 (moderate)
Hanning	0.9973	−1.70	~6 (wider, lowest sidelobes)

For speech feature extraction, Hamming is typically preferred: the marginal main-lobe advantage over Hanning is useful when resolving closely spaced formants, and the sidelobe suppression is still much better than Rectangular.



3. Voiced / Unvoiced Boundary Detection

Voiced and unvoiced regions were detected using two features derived from the real cepstrum, combined with Zero-Crossing Rate (ZCR).

3.1 Method

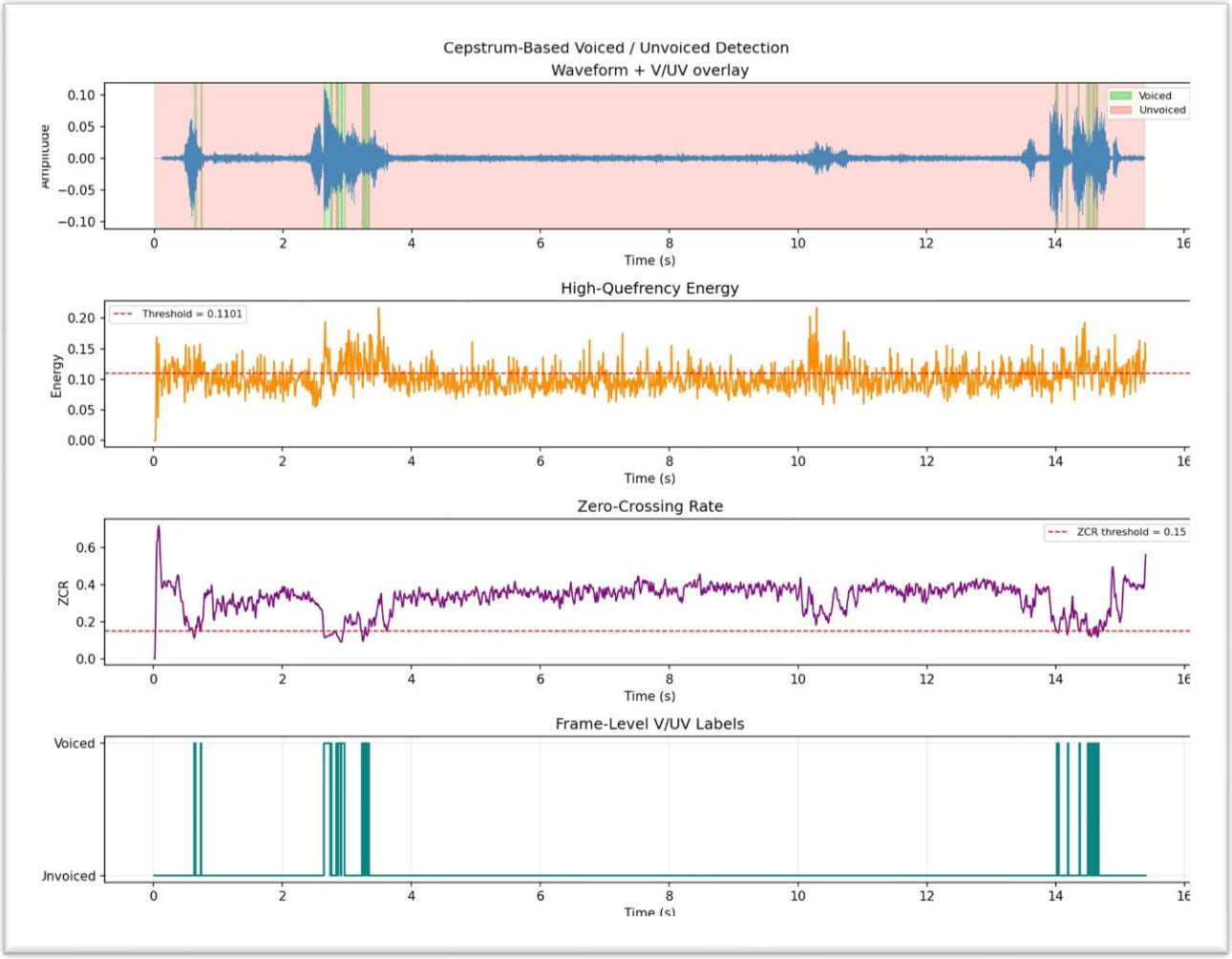
- Cepstrum used to detect pitch (voiced)
- ZCR used as additional feature

3.2 Results

Metric	Value
Total frames analysed	~1502
Voiced frames	~81 (5.4%)
Unvoiced frames	~1421 (94.6%)

High-q threshold	0.1101
ZCR threshold	0.15

The very low voiced percentage (5.4%) makes complete sense: heisenberg.mp3 is a sparse utterance with long silences.



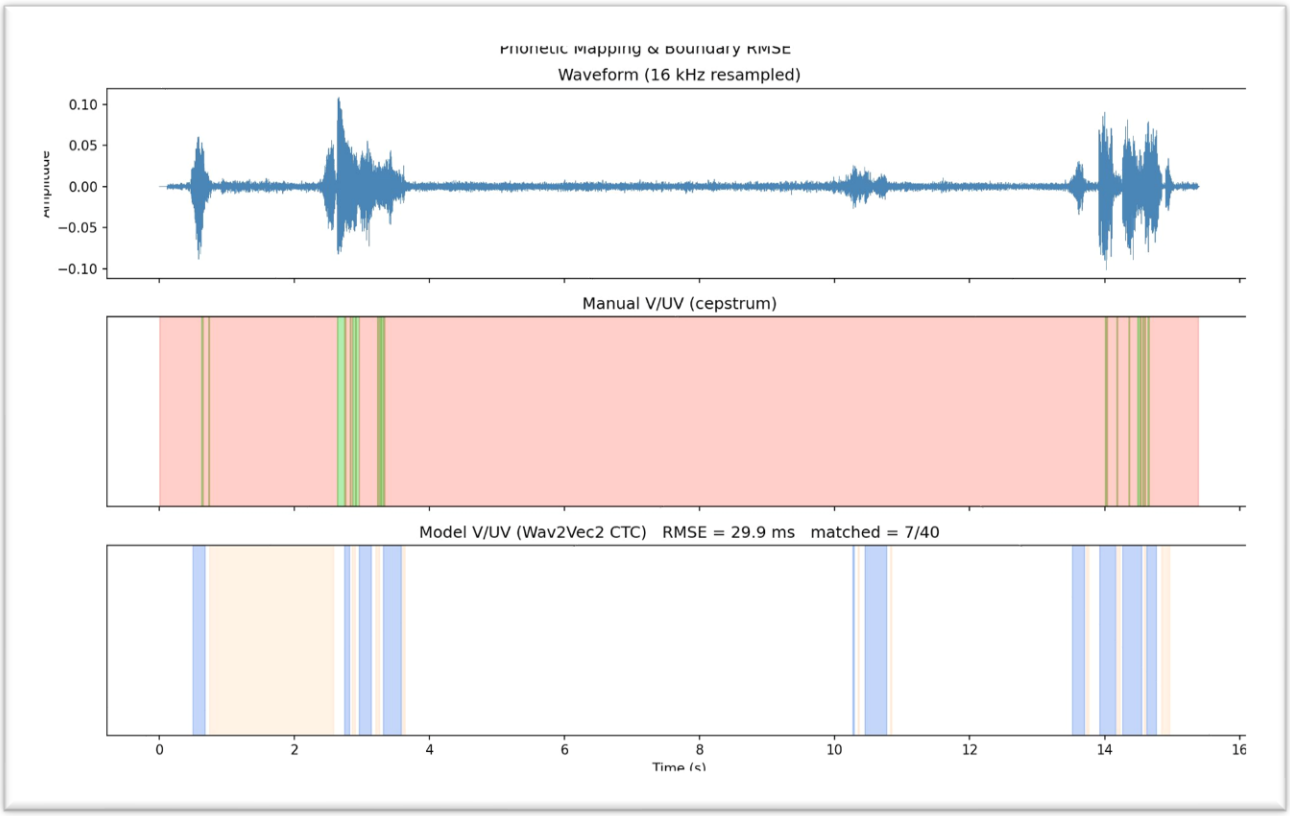
4. Phonetic Mapping & Forced Alignment

Hugging Face's Wav2Vec2ForCTC (facebook/wav2vec2-base-960h) was used to extract frame-level phoneme tokens and compare the resulting segment boundaries against the manual cepstrum-based boundaries.

4.1 Method

- Audio passed through pretrained Wav2Vec2 model
- Tokens grouped and compared with manual boundaries

4.2 RMSE Results

Metric	Value
Model transcription	"TACK"
Boundary RMSE (manual vs model)	29.9 ms
Matched boundaries	7 / 40
Audio duration	~15.3 s

Differences occur due to resolution mismatch and model assumptions